ANALYTICAL CUMULATIVE EXAM

September 16, 1995

Transferrin is the iron-transport protein found in blood. It has a molecular weight of 81,000 and carries two Fe^{3+} ions (atomic weight of iron is 55.8). Desferrioxamine B is a potent iron chelator used to treat patients with iron overload. It has a molecular weight of about 650 and can bind one Fe^{3+}. Desferrioxamine can take iron from many sites within the body and is excreted (with its iron) through the kidneys. The molar absorptivities of these two compounds (saturated with iron) at two wavelengths are given below. Both compounds are colorless in the absence of iron.

λ	ϵ (M^{-1} cm^{-1}) Transferrin	ϵ (M^{-1} cm^{-1}) Desferrioxamine
428	3540	2730
470	4170	2290

1) (50 points) A solution of transferrin exhibits an absorbance of 0.463 at 470 nm in a 1.000 cm cell. a) Calculate the concentration of transferrin in milligrams per milliliter, and b) the concentration of iron in micrograms per milliliter.

2) (50 points) A sample containing both transferrin and desferrioxamine is analyzed showing an absorbance of 0.424 at 470 nm and 0.401 at 428 nm. a) what is the concentration of transferrin and desferrioxamine in the sample? b) Calculate the fraction of iron in transferrin and the fraction of iron in desferrioxamine. Remember that transferrin binds with two irons and desferrioxamine just one.